

READING PASSAGE 2

You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.

Nanotechnology: the science of the very small

- A** Nanoscience and nanotechnology have become very fashionable in recent times. Their prominence has inspired science fiction and raised a lot of hope and hype about the possibilities. But what exactly is nanotechnology? Most definitions agree that it is the manufacture of structures that measure up to 100 nanometres (nm). The diameter of an atom is typically only two or three tenths of a nanometre while that of the human hair is around 1,000,000 nm. Essentially, we are now able to build new structures from atoms and molecules as if we were using the bricks of a children's construction toy.
- B** Most of the electronic consumer goods that we use today contain components that measure less than 100 nm. The lasers in our CD and DVD players make use of the properties of semiconductor layers that are just a few atoms in thickness, so we clearly are in the nanotechnology age. There are many other everyday items that we take for granted that have nanoparticles in them. One of these is sunscreen. In the 1990s there were safety issues with many of the sunscreen products available, so scientists at Oxford University set about finding a solution. The product they developed uses nanoparticles of titanium dioxide that are doped with the element manganese. This product is almost transparent to the eye when spread on the skin, and it is effective for much longer than many of the traditional preparations. In this case, we see an example of nanotechnology providing a solution to an existing problem rather than opening up new product areas.
- C** The technique of using nanotechnology to adapt existing products can also be applied to garments, upholstery, paint, and plastics to improve their longevity. Bearing this in mind, it is interesting to ask whether nanotechnology can help with some of the world's major current issues. The recent growth in climate change awareness calls a range of things into question. Can we use energy more effectively? Can we reduce the carbon dioxide in the atmosphere? Can we use and purify water more effectively? Can health care be improved? Nanotechnology may very well provide the solutions. There are few times in history when the application of science to improving the human condition has been so apparent.
- D** We are already making new types of solar cells and batteries by using our newfound ability to manipulate matter at the atomic and molecular scale. If we can capture some of the huge quantities of energy that arrive at the Earth's surface from the Sun, and store it effectively, we can reduce the need for fossil fuel-based power stations. We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases, and we may possibly convert some of them back into fuel. There are, however,

many technical problems involved, and it will be another decade or so before some of these ideas are deployed.

- E** Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil. Currently we do not use water efficiently, and many industrial and food processes leave contamination in water that is potentially toxic and difficult to remove. Nanofiltration may provide the solution to these problems, and may also offer new routes to the desalination of brackish water and groundwater in some parts of the world.
- F** Food production and processing are becoming more important as the world's population increases. There are concerns that potentially harmful nano-products might enter the body via the gut. Such concerns are being addressed, but it is even more important to ask questions about how nanotechnology might increase the nutritious yield of food and be adapted to help preserve and package food. There is already evidence to show that it might also be helpful in the primary production of raw food materials, by way of better delivery of pesticides in the field and better extraction of protein and carbohydrates.
- G** Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago. It is happening across several fronts: better appreciation of what the 'natural' nanoparticles circulating in body fluid do; better methods of delivering drugs to the body via nanoparticles that target the diseased body part; improvements to the diagnostic imaging techniques of ultrasound, magnetic resonance imaging and X-ray computerised tomography. We are now at the threshold of a new subject, 'theranostics', whereby nanoparticles will first deliver the diagnostic information to the clinician and then, using the same particle, deliver a form of therapy.
- H** Nanotechnology is a life-changing technology, making the quality of life better, enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities. There are many other examples in the world of 'new materials' that also make use of nanotechnology. The possibilities may well be limitless.

Questions 14–16

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 14–16 on your answer sheet.

Nanotechnology

In the same way that a child may use **14**, scientists are now able to create new structures by combining the smallest particles known to man. We can fully appreciate the microscopic scale of this work by comparing the exact **15** of one atom with that of a single human hair.

A lot of today's electronic consumer goods, such as lasers, have components that are microscopic in size. They are not the only products relying on nanotechnology. For example, a long-lasting **16**, widely available today, relies on minute nanoparticles for its effectiveness. This was developed in the 1990s as a result of concerns about the safety of existing products.

Questions 17–22

Reading Passage 2 has eight paragraphs, **A–H**.

Which paragraph contains the following information?

Write the correct letter, **A–H**, in boxes 17–22 on your answer sheet.

- 17** a comparison between two of the world's vital natural resources
- 18** a mention of how nanotechnology could be dangerous for humans
- 19** a claim that nanotechnology has influenced modern authors
- 20** a reference to the fact that nanotechnology is achieving more than was anticipated a decade ago
- 21** a suggestion that nanotechnology could provide financial as well as ecological benefits
- 22** an acknowledgement that more time is required to realise some of nanotechnology's potential

Questions 23 and 24

Choose **TWO** letters, **A–E**.

Write the correct letters in boxes 23 and 24 on your answer sheet.

Which **TWO** of the following statements does the writer make about nanotechnology and energy resources?

- A** Nanotechnology means that solar energy will replace fossil fuel energy.
- B** We have made solar batteries for the first time by applying nanotechnology.
- C** Further research is needed to turn nanotechnology into practical energy solutions.
- D** Nanotechnology is being used to remove dangerous gases from the atmosphere.
- E** Nanotechnology has helped create cheaper sources of energy.

Questions 25 and 26

Choose **TWO** letters, **A–E**.

Write the correct letters in boxes 25 and 26 on your answer sheet.

In which **TWO** ways does the writer think nanotechnology could improve food production?

- A** by enabling crops to grow faster
- B** by keeping food fresh for longer
- C** by making food tastier
- D** by protecting food from insects better
- E** by increasing the variety of food

Disclaimer

Compiled, formatted, and lightly proofread by ZYZ Reading Walks.

All copyright in the underlying works remains with the original authors and publishers.

No affiliation with or endorsement by any rights holder (including IELTS® owners).

For non-commercial educational use only. This notice must remain intact in all copies.

Available free of charge from ZYZ Reading Walks. Resale or any paid distribution is prohibited.

Questions 14–16 填空题

题号	答案	题干翻译	详细定位与翻译	详细解释
14	bricks	就像孩子可以使用 14..... 一样，科学家现在能够通过组合人类已知的最小粒子来创造新结构。	第 A 段： <i>Essentially, we are now able to build new structures from atoms and molecules as if we were using the bricks of a children's construction toy.</i> 译：本质上，我们现在能够用原子和分子建造新结构，就好像我们在使用儿童搭建玩具中的积木一样。	题干中的 <i>In the same way that a child may use...</i> 对应原文中的 <i>as if we were using...</i> ；题干中的 <i>create new structures by combining the smallest particles</i> 对应 <i>build new structures from atoms and molecules</i> 。空格处需要填孩子用来搭建东西的物品，所以答案是 bricks 。
15	diameter	通过比较一个原子的确切 15..... 和一根人类头发的相应数值，我们可以充分理解这种工作的微观尺度。	第 A 段： <i>The diameter of an atom is typically only two or three tenths of a nanometre while that of the human hair is around 100,000 nm.</i> 译：一个原子的直径通常只有十分之二或十分之三纳米，而一根人类头发的直径大约是 100,000 纳米。	题干中的 <i>comparing the exact ... of one atom with that of a single human hair</i> 对应原文 <i>The diameter of an atom... while that of the human hair...</i> 。这里比较的是原子和头发的直径，所以答案是 diameter 。
16	sunscreen	例如，如今广泛可得、效果持久的 16..... 依靠微小纳米颗粒发挥作用。它是在 20 世纪 90 年代由于人们担心现有产品的安全性而开发出来的。	第 B 段： <i>One of these is sunscreen. In the 1990s there were safety issues with many of the sunscreen products available... The product they developed uses nanoparticles of titanium dioxide... This product is almost transparent... and it is effective for much longer...</i> 译：其中之一是防晒霜。20 世纪 90 年代，当时许多可获得的防晒产品存在安全问题.....他们开发的产品使用了二氧化钛纳米颗粒.....这种产品涂在皮肤上几乎是透明的，而且比许多传统制剂有效时间更长。	题干中的 <i>long-lasting</i> 对应原文 <i>effective for much longer</i> ； <i>widely available today</i> 对应原文提到的 <i>sunscreen products available</i> ； <i>relies on minute nanoparticles</i> 对应 <i>uses nanoparticles of titanium dioxide</i> 。所以答案是 sunscreen 。

Questions 17–22 段落信息匹配题

题号	答案	题干翻译	详细定位与翻译	详细解释
17	E	对世界上两种重要自然资源的比较	第 E 段： <i>Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil.</i> 译：水净化正变得越来越具有挑战性，许多人认为，我们习以为常的水将会变得比石油更加珍贵。	题干中的 <i>two of the world's vital natural resources</i> 指的是 water 和 oil 。原文直接比较了水和石油的价值：水可能会比石油更有价值。因此答案是 E 。
18	F	提到纳米技术可能如何对人类造成危险	第 F 段： <i>There are concerns that potentially harmful nano-products might enter the body via the gut.</i> 译：人们担心潜在有害的纳米产品可能会通过肠道进入人体。	题干中的 <i>could be dangerous for humans</i> 对应原文 <i>potentially harmful nano-products might enter the body</i> 。这里明确说纳米产品可能通过肠道进入人体，因此构成健康风险。答案是 F 。
19	A	声称纳米技术影响了现代作家	第 A 段： <i>Their prominence has inspired science fiction and raised a lot of hope and hype about the possibilities.</i> 译：它们的突出地位激发了科幻小说创作，并引发了人们对其可能性的许多希望和炒作。	题干中的 <i>modern authors</i> 被原文具体化为 <i>science fiction</i> ，也就是科幻文学/科幻小说作者。 <i>has influenced</i> 对应 <i>has inspired</i> 。所以答案是 A 。
20	G	提到纳米技术正在取得超过十年前预期的成果	第 G 段： <i>Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago.</i> 译：医疗保健是一个纳米技术应用已经超过十年前预期的领域。	题干中的 <i>achieving more than was anticipated a decade ago</i> 几乎是原文 <i>exceeding the expectations of ten years ago</i> 的同义改写。 <i>a decade ago = ten years ago</i> 。答案是 G 。

21	H	暗示纳米技术可以带来经济利益和生态利益	第H段: <i>Nanotechnology is a life-changing technology, making the quality of life better, enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities.</i> 译: 纳米技术是一种改变生活的技术, 它能改善生活质量, 使我们更有效地利用资源, 从而保护环境, 并创造新的商业机会。	题干中的 <i>ecological benefits</i> 对应 <i>protecting our environment</i> ; <i>financial benefits</i> 对应 <i>creating new business opportunities</i> 。同一句中同时出现环境保护和商业机会, 所以答案是H。
22	D	承认实现纳米技术的某些潜力还需要更多时间	第D段: <i>There are, however, many technical problems involved, and it will be another decade or so before some of these ideas are deployed.</i> 译: 然而, 其中涉及许多技术问题, 有些想法还需要大约十年时间才能被实际应用。	题干中的 <i>more time is required</i> 对应原文 <i>it will be another decade or so before...</i> ; <i>realise some of nanotechnology's potential</i> 对应 <i>some of these ideas are deployed</i> 。这里强调不是现在马上实现, 而是仍需时间和技术突破。答案是D。

Questions 23–24 多选题: 纳米技术与能源资源

正确答案: C、D

选项	是否正确	题干翻译	详细定位与翻译	详细解释
A	×	纳米技术意味着太阳能将取代化石燃料能源。	第D段: <i>If we can capture some of the huge quantities of energy... we can reduce the need for fossil fuel-based power stations.</i> 译: 如果我们能够捕获一部分到达地球表面的巨大太阳能, 并有效储存, 就能减少对化石燃料发电站的需求。	原文只是说 reduce the need for fossil fuel-based power stations , 即“减少需求”, 不是“完全取代”。 <i>will replace</i> 太绝对, 原文没有支持。
B	×	我们通过应用纳米技术第一次制造出了太阳能电池。	第D段: <i>We are already making new types of solar cells and batteries by using our newfound ability...</i> 译: 我们已经在利用新获得的能力制造新型太阳能电池和电池。	原文说的是 new types of solar cells and batteries , 不是“第一次制造太阳能电池”。 <i>for the first time</i> 属于无中生有。
C	✓	还需要进一步研究, 才能把纳米技术转化为实际能源解决方案。	第D段: <i>There are, however, many technical problems involved, and it will be another decade or so before some of these ideas are deployed.</i> 译: 然而, 其中涉及许多技术问题, 有些想法还需要大约十年时间才能被实际应用。	原文的 <i>many technical problems</i> 和 <i>another decade or so before... deployed</i> 说明这些能源想法还不能马上实际应用, 需要进一步研究和时间。因此C正确。
D	✓	纳米技术正在被用于从大气中去除危险气体。	第D段: <i>We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases...</i> 译: 我们已经在使用纳米颗粒和纳米多孔材料来捕获有害温室气体。	<i>dangerous gases from the atmosphere</i> 对应原文 <i>harmful greenhouse gases</i> 。虽然原文说的是 <i>capture</i> , 题干说 <i>remove</i> , 但语义一致: 捕获这些有害气体, 使其不继续留在大气中造成危害。因此D正确。
E	×	纳米技术已经帮助创造了更便宜的能源来源。	第D段没有提到 <i>cheaper sources of energy</i> 。	原文说到了太阳能捕获、储存、减少化石燃料需求、捕获温室气体, 但没有说能源变得更便宜。 <i>cheaper</i> 无对应信息。

Questions 25–26 多选题：纳米技术如何改善食品生产

正确答案：B、D

选项	是否正确	题干翻译	详细定位与翻译	详细解释
A	×	使农作物生长得更快	第F段没有提到 <i>grow faster</i> 。	原文提到提高营养产量、保存和包装食物、农药输送、蛋白质和碳水化合物提取，但没有说让作物更快生长。
B	✓	让食物保持新鲜更久	第F段：... <i>how nanotechnology might increase the nutritious yield of food and be adapted to help preserve and package food</i> . 译：.....纳米技术如何可能提高食物的营养产量，并被改造用于帮助保存和包装食物。	<i>keeping food fresh for longer</i> 对应原文 <i>preserve and package food</i> 。保存和包装食物的目的就是延长食物保鲜时间，所以 B 正确。
C	×	让食物更美味	第F段没有提到味道。	原文没有出现 <i>taste / flavour / tastier</i> 等信息。 <i>making food tastier</i> 属于无中生有。
D	✓	更好地保护食物免受昆虫侵害	第F段：... <i>helpful in the primary production of raw food materials, by way of better delivery of pesticides in the field...</i> 译：.....通过在田间更好地输送农药，它也可能有助于原始食物材料的初级生产。	<i>better delivery of pesticides</i> 指更有效地使用农药，而农药的作用通常包括防治害虫。因此题干中的 <i>protecting food from insects better</i> 是合理同义改写。D 正确。
E	×	增加食物种类	第F段： <i>increase the nutritious yield of food</i> 译：提高食物的营养产量。	很容易误选 E，但 <i>yield</i> 是“产量”，不是 <i>variety</i> “种类”。原文说的是提高营养产量和改善生产、保存方式，不是增加食物种类。